

→ Basic program in **JAVA** :-

Q. WAP to print "hello World".
 ⇒

```
class pogo
{
    public static void main (String[] args)
    {
        System.out.println("hello World");
    }
}
```

08

```
class ---
{
    public ---
    {
        System.out.print("hello World");
    }
}
```

→ How to move in **next line** ?

⇒

```
System.out.print("hello World");
System.out.print("Akash Kumar");
```

Output → hello WorldAkash Kumar

We use :

```
System.out.println("hello. World");
System.out.print("Akash kumar");
```

output → hellow World
 Akash kumar

{ ~~print~~ln → Used for next line }

OR

```
System.out.print("hello World\n");
System.out.print("Akash kumar");
```

Output → hello World
 Akash kumar.

```
public static void main (String[] args)
```

OR

```
" " " " (String args[])
```

OR

```
" " " " (String... args)
```

→ Variables and their declaration :-

```
int x; // declaration  
x = 5; // initialization
```

OR

```
int x = 5; // declaration & initialization.
```

~~Example~~

```
int x;  
System.out.print(x);
```

output = error { initialization }

```
int x;
```

```
x = 6;
```

```
System.out.print(x);
```

output = 6.

→ Arithmetic operation:~

int x = 25;

int y = 10;

S.o.p (x+y);

S.o.p (x-y);

S.o.p (x*y);

S.o.p (x/y); // 2.5

Output → ~~35~~ 35

15

250

2

// wrong output.

We use float / double

float x = 25f; or x = 25

int y = 10;

S.o.p (x/y);

2.5

[we use f when
 we have to
 write in floats
 Eg:- 25.5f ✓
 x = 25.5f ✗]

Q. WAP to print area of circle?

```
double r = 7.5;
double a = 3.14159 * r * r;
S.o.p(a);
```

Output = 176.7144375

OR

```
double r = 7.5;
double a = 3.14159 * r * r;
S.o.p("Area of circle:" + a);
```

Output: Area of circle: 176.714475

OR

```
double r = 7.5;
double a = 3.14159 * r * r;
S.o.p("Area of circle with radius: " + r + " is: " + a);
```

Output:- Area of circle with radius: 7.5 is: 176.7144375.

Q. WAP to print volume of a sphere?

double

double r = 7.5;
 double v = $\frac{4}{3} * 3.14159 * r * r * r$;
 S.o.p (v); // 1767.14437

declare as integer
 so treated as int.
 (4/3 = 1)

Output => 1325.35 // wrong

double v = ~~(4 * 3.1415 * r * r * r)~~ / 3

Output = 1767.14437

or

double v = $\frac{4.0}{3} * 3.14159 * r * r * r$

output = 1767.14437

~~declaration~~ declared as
 double / float.

→ Variable Naming rules :-

1. Variables can start from an alphabet or underscore _ or \$.
2. Special character except _ and \$ are not allowed.
3. Some particular keywords are not allowed.
4. Commas or blanks are not allowed.

→ Taking a input:~

```
import java.util.Scanner;
```

```
class Pogo  
{
```

```
    public static void main(String[], args)  
    {
```

```
        Scanner sc = new Scanner(System.in);  
        System.out.println("enter a value");
```

```
        int a = sc.nextInt();
```

```
        S.o.p("square of a num=" + a*a);
```

```
    }
```

```
}
```

→ Take input of two num & print Sum :-

```
import java.util.Scanner;
```

```
class pogo  
{
```

```
    public static void main(String[], args)  
    {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Enter two num");
```

```
        int a = sc.nextInt();
```

```
        int b = sc.nextInt();
```

```
        System.out.println("Sum of num = " + (a+b));  
    }
```

```
}
```

→ Conditionals :-

Q. Take positive integer input and tell if it is even or odd.

⇒

```
import java.util.Scanner;
```

```
class prog01 {
```

```
    public static void main (String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Enter a num");
```

```
        int a = sc.nextInt();
```

```
        if(a % 2 == 0)
```

```
            System.out.println("even");
```

```
        else
```

```
            System.out.println("odd");
```

```
    }
```

```
}
```

Q. Take positive integers input and tells if it is divisible by 5 or not.

```
import java.util.Scanner;
```

```
class Pogo
```

```
{
```

```
    public static void main (String[] args)
```

```
    {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        Sout("enter a no.");
```

```
        int a = sc.nextInt();
```

```
        if (a % 5 == 0)
```

```
            sout("Yes");
```

```
        else
```

```
            sout("No");
```

```
    }
```

```
}
```

Q. Take integer input & print the absolute value of that integer.

=>

$n = 10 \rightarrow 10$
 $n = -4 \rightarrow 4$
 $n = -6 \rightarrow 6$

In maths we called mod
 $| -4 | = 4$
 $| -6 | = 6$

$\rightarrow$ import java.util.Scanner;

class pogo {

public static void main(String... args)

{
Scanner sc = new Scanner(System.in);

scout("enter a no.");

int a = sc.nextInt();

~~if(a % 2 == 0)~~

~~scout("Yes");~~

~~else~~

~~scout("No");~~

if(a < 0)

scout(-a);

else

scout(a);

}

}

Q. If cost price and selling price of an item is input through the keyboard, write a program to determine whether a seller has made profit or loss or no profit no loss. Also determine how much profit he made or loss he incurred.

```
import java.util.Scanner;
```

```
class Pogo {
```

```
    public static void main (String... args)
```

```
    {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        Sout("Enter CP & SP");
```

```
        int CP = sc.nextInt();
```

```
        int SP = sc.nextInt();
```

```
        if (CP > SP)
```

```
            Sout("loss of " + (CP - SP));
```

```
        else-if (CP == SP)
```

```
            Sout("no profit no loss");
```

```
        else
```

```
            Sout("profit of " + (SP - CP));
```

```
    }
```

```
}
```

Q. Take positive integer input ^{and} tell if it is a three digit number or not.

→

```
import java.util.Scanner;
```

```
class Pogo  
{
```

```
    public static void main (String... args)  
    {
```

```
        Scanner sc = new Scanner (System.in);
```

```
        Sout ("enter a no.");
```

```
        int a = sc.nextInt();
```

```
        if (a > 99 && a < 1000)
```

```
            sout ("three digit no.");
```

```
        else
```

```
            sout ("not a three digit no.");
```

```
    }
```

```
}
```

Q. take a positive integer input and tell if it is divisible by ^{both} 5 and 3.

⇒

```
import java.util.Scanner;
class Pogo {
    public static void main(String... args) {
        Scanner sc = new Scanner(System.in);
        Sout("enter a no.");
        int a = sc.nextInt();
        if((a%3 == 0) && (a%5 == 0))
            Sout("divisible by both 5 & 3);
        else
            Sout("not divisible by both 5 & 3);
    }
}
```

Q take a positive integer input and tell if it is divisible by 5 or 3.

```
import java.util.Scanner;
class Pogo {
    public static void main(String... args) {
        Scanner sc = new Scanner(System.in);
        Sout("enter a num");
        int a = sc.nextInt();
        if((a%3 == 0) || (a%5 == 0))
            Sout("divisible by both 3 or 5);
        else
            Sout("not divisible by 3 or 5);
    }
}
```

Q take 3 positive integer input and print the greatest of them.

⇒

```
import java.util. Scanner;  
class pogo {  
    public static void main(String... args) {  
        Scanner sc = new Scanner (System.in);  
        Sout("enter three no.");  
        int a = sc.nextInt();  
        int b = sc.nextInt();  
        int c = sc.nextInt();  
  
        if( a > b && a > c)  
            Sout(a + " is largest no.");  
        else if( b > c)  
            sout(b + " is largest- no.");  
        else  
            sout("c + " is largest- no.");  
    }  
}
```

Q. Take input percentage of a student and print the grade according to marks:-

- 1.) 81-100 Very good
- 2.) 61-80 Good
- 3.) 41-60 Average
- 4.) ≤ 40 Fail

```
import java.util.Scanner;
class pogo {
    public static void main(String... args) {
        Scanner sc = new Scanner(System.in);
        sout("enter a percentage");
        int a = sc.nextInt();

        if(a > 80 && a <= 100)
            sout("Very Good");
        else if(a > 60 && a <= 80)
            sout("Good");
        else if(a > 40 && a <= 60)
            sout("Average");
        else if(a >= 0 && a <= 40)
            sout("Fail");
        else
            sout("Invalid score");
    }
}
```

Q. Given a point (x, y) , write a program to find out if lies in the 1st quadrant, 2nd quadrant, 3rd quadrant, 4th quadrant, On the x -axis, y -axis, or at the origin, viz $(0, 0)$.

⇒

```

if (x == 0 && y == 0)
    cout ("at origin");

else if ((x > 0 || x < 0) && y == 0)
    cout ("on x-axis");

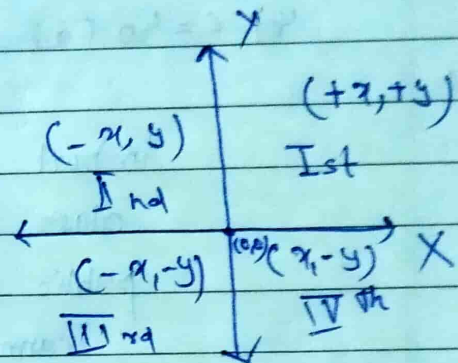
else if (x == 0 && (y > 0 || y < 0))
    cout ("on y-axis");

else if (x > 0 && x y > 0)
    cout ("1st quadrant");

else if (x < 0 && y > 0)
    cout ("2nd quad");

else if (x < 0 && y < 0)
    cout ("3rd quad");

else if (x > 0 && y < 0)
    cout ("4th quad");
    
```



→ Ternary operator :- { Replacement for if-else :-

Expression ? True : False

Eg;

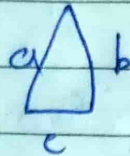
$(x \% 2 == 0) ? \text{"Even"} : \text{"odd"}$

`cout (  $x \% 2 == 0 ? \text{"Even"} : \text{"odd"}$`);

Q. Take 3 numbers input and tell if they can be the side of a triangle.

⇒

$a, b, c \rightarrow \text{side}$



$$a + b > c$$

$$b + c > a$$

$$a + c > b$$

```
if ((a+b) > c && (b+c) > a && (a+c) > b)
    cout << "triangle possible";
```

else

```
    cout << "NOT possible";
```

→ Switch Statement & Replacement of if-else :-

Syntax :-

```
Switch (constant expression) {  
    case const. 1 :  
        // body  
    case const. 2 :  
        // body  
        :  
    default :  
        // body  
}
```

Q. WAP to create a ~~calculator~~ calculator (+, -, /, *) :-

```
Scout("enter two num");  
int a = sc.nextInt();  
int b = sc.nextInt();  
Scout("enter operator (+, -, *, /)");  
char operator = sc.next().charAt(0);  
Switch (operator) {  
    case '+':  
        Scout(a+b);  
        break;  
        :  
    default:  
        Scout("error");  
}
```

$$y = ax^2$$

$$y = 4x^2$$

→ Loops :-

Q. Print hello world 'n' times. Takes 'n' as input from user.

⇒

```
import java.util.Scanner;
class pogo {
    public static void main(String... args)
    {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
```

```
        for(int i=0; i<n; i++)
            System.out.println("hello world");
    }
}
```

→ How for loop works :- The various parameter

Initialization Condition increment

```
for(i=0; i<5; i++)
{
    // body
}
```

Steps:-

i = 0.
 i < 5, 0 < 5 ✓
 // body
 increment

i = 1 ... 2, 3, 4, 5

i = 5, 5 < 5 X
 loop breaks

Initialization

condition check

Go inside loop (body)

increment

Repeat

Q. Print number from 1 to 100.

⇒

```
import java.util.Scanner;
```

```
class pogo {
```

```
    public static void main(String... args) {
```

```
        for(int i = 1; i <= 100; i++)
```

```
            sout(i);
```

```
        }
```

```
    }
```

Q. Print all the even no. from 1 to 100.

⇒

```
import java.util.Scanner;
```

```
class pogo {
```

```
    public static void main(String... args) {
```

```
        for(int i = 1; i <= 100; i++) {
```

```
            if(i % 2 == 0)
```

```
                sout(i);
```

```
        }
```

```
    }
```

```
}
```

Q. Print all the odd number from 1 to 100.

```
import java.util.Scanner;

class pogo {

    public static void main(String... args) {

        for(int i=1; i<=100; i++) {

            if(i%2 != 0)

                sout(i);

        }

    }

}
```

Q. Print the table of 19.

```
import java.util.Scanner;

class pogo {

    public static void main(String... args) {

        for(i=1; i<=10; i++)

            Sout(i*19);

    }

}
```

Q. Print the table of 'n'. Here 'n' is a integer which user will input.

⇒

```
import java.util. Scanner;
```

```
class pogo {
```

```
public static void main (String... args) {
```

```
Scanner sc = new Scanner (System.in);
```

```
Scanner sc = new Scanner (System.in);
```

```
    Sout("enter a no.");
```

```
    int x = sc.nextInt();
```

```
    for(int i = 1; i <= 10; i++)
```

```
        Sout(i * x);
```

```
    }
```

```
}
```

Q. Display this AP (arithmetic progression) -

1, 3, 5, 7, 9, ... n terms.

⇒

```
import java.util.Scanner;
```

```
class pogo {
```

```
public static void main(String... args) {
```

```
Scanner sc = new new Scanner(System.in);
```

```
    Sout("enter a no.");
```

```
    int n = sc.nextInt();
```

```
    for(i=1; i<=n; i++ i+=2)
```

```
        Sout(i);
```

```
    }
```

```
}
```

Q. Display this AP - 4, 7, 10, 13, 16, ... n terms.

```
for(i=4; i<=n; i+=3)
```

```
    sout(i);
```

Or

```
int a=4;
```

```
for(i=1; i<=n; i++) {
```

```
    int a=4; int a=4; int a=4;
```

```
    sout(a); sout(a);
```

```
}
```

```
    Sout(a);
```

```
    a+=3;
```

→ Q. Display this GP - 1, 2, 4, 8, 16, 32, ... n terms.

```
for(int i=1; i<=n; i*=2)
    cout<<i<<endl;
```

OR

```
int a=1;
for(int i=1; i<=n; i++) {
    cout<<a<<endl;
    a*=2;
}
```

3

→

Break:-

↳ keyword used to finish / terminate a loop.

Q. WAP to check a prime no. or not.

⇒

Eg: 1 (is not a prime no.)

36 → 1, 2, 3, 4, 6, 9, 12, 18, 36
 (Not a prime no.)

23 → 1, 23
 (prime no.)

(45) → 1, 3, 5, 9, 15, 45
 Not a prime no.)

if a number is not divisibly from 2 to $n/2$
 then it is a prime no. → Time → $O(n)$

We can also use 2 to $\sqrt{n}$ also
 ↳ $O(\sqrt{n})$

```
import java.util.Scanner;
class Pogo {
    public static void main(String... args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        Sout("enter a number");
```

```
        int count = 0;
```

```
        int n = sc.nextInt();
```

```
        if (n <= 1)
```

```
            Sout("not a prime no.");
```

```
        else
```

```
        {
```

```
            for (int i = 2; i <= n/2; i++) // i <= Math.sqrt(n)
            {
```

```
                if (n % i == 0)
```

```
                {
```

```
                    count++;
```

```
                    break;
```

```
                }
```

```
            }
```

```
            if (count == 1)
```

```
                Sout("not a prime no.");
```

```
            else
```

```
                Sout("prime no.");
```

```
        }
```

```
    }
```

```
}
```

→ Continue :-

↳ kuch particular iteration ko skip kardo

```
import java.util.Scanner;
```

```
class pogo {
```

```
public static void main (String... args) {
```

```
    for (int i = 0; i < 10; i++) {
```

```
        if (i == 3)
```

```
            continue;
```

```
        cout(i);
```

```
    }
```

```
}
```

```
}
```

output :-

0

1

2

4

5

6

7

8

9

→ While loop is replacement of for loop.

```
int i = 0;
while(i < 10) {
    // body
    i++;
}
```

→ Do-while loop is - Useless

```
do
{
    // body
}
while (another == 'y');
```

→ Pattern printing :-

Q. Print the given patterns :-

```
* * * *
* * * *
* * * *
* * * *
```

⇒

```
import java.util.Scanner;
```

```
class pogo {
```

```
public static void main(String... args) {
```

```
    for (int i=0; i<4; i++) {
```

```
        for (int j=0; j<4; j++) {
```

```
            sout(" * "); // don't use println, use print only
```

```
        }
```

```
        sout("\n"); // or sout.println();
```

```
    }
```

```
}
```

```
}
```

Q. Print the given pattern :-

```
1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4
```

```

for (int i = 1; i <= 4; i++) {
    for (int j = 1; j <= 4; j++) {
        cout (j + " ");
    }
    cout ("\\n");
}

```

Q. Print the given pattern:-

```

A B C D
A B C D
A B C D
A B C D

```

⇒

Method 1:-

```

for (int i = 1; i <= 4; i++)
{
    for (char j = 65; j <= 68; j++)
    {
        cout (j + " ");
    }
    cout ("\\n");
}

```

Method 2:-

```

for (int i = 1; i <= 4; i++)
{
    for (j = 1; j <= 4; j++)
    {
        cout (char) (j + 64) + " ";
    }
    cout ("\\n");
}

```

Q. Print the following pattern:-

```
A A A A
B B B B
C C C C
D D D D
```

⇒

```
for (int i = 1; i <= 4; i++)
{
    for (int j = 1; j <= 4; j++)
    {
        cout << (char) (i + 64) << " ";
    }
    cout << "\n";
}
```

Q. Print the given pattern

```
*
* *
* * *
* * * *
```

⇒

```
for (int i = 1; i <= 4; i++) {
    for (int j = 1; j <= i; j++) {
        cout << "*" << " ";
    }
    cout << "\n";
}
```

Q. Print the given pattern:-

```
1
1 2
1 2 3
1 2 3 4
```

→

```
for(int i=1; i<=4; i++)
{
    for(int j=1; j<=i; j++)
    {
        cout<<j<<" ";
    }
    cout<<"\n";
}
```

Q. Print the given pattern:-

```
A
A B
A B C
A B C D
```

→

```
for(int i=1; i<=4; i++) {
    for(int j=1; j<=i; j++) {
        cout<<(char)(j+64)<<" ";
    }
    cout<<"\n";
}
```

Q. Print the given pattern:

```

1
2 2
3 3 3
4 4 4 4
    
```

→

```

for (int i = 1; i <= 4; i++)
{
    for (int j = 1; j <= i; j++)
    {
        cout << " ";
    }
    cout << "\n";
}
    
```

Q. Print the given pattern:-

```

1
A B
1 2 3
A B C D
    
```

→

```

for (int i = 1; i <= 4; i++)
{
    for (int j = 1; j <= i; j++)
    {
        if (i % 2 == 0)
            cout << (char)(i + 64) << " ";
        else
            cout << j << " ";
    }
    cout << "\n";
}
    
```

1
2 3
4 5 6
7 8 9 10
MON TUE WED THU FRI SAT SUN

int a = 1;
{ same as before;
sout(a++);
}

Page No.: _____

Date: ____/____/____

Q. Print the given pattern

* * * *
* * *
* *
*

for (int i = 1; i <= 4; i++)

{

for (int j = 4; j >= i; j--) // int j = 1; j <= 4 + 1 - i; j++
{

sout(" * ");

}

sout("\n");

}

Q. Print the given pattern:-

1 2 5 7
1 3 5
1 3
1

for (int i = 1; i <= 7; i = i + 2)

{

for (j = 1; j <= 7 + 1 - i; j = j + 2)

{

sout(j + " ");

}

sout("\n");